Slide 1: Title

Good morning/afternoon everyone.

I am Devi Lalitha from the Department of Computer Science and Engineering (Data Science), PSCMR College of Engineering and Technology.

Today, I am presenting our research paper titled:

"AI-Based Healthcare Data Analysis for Early Disease Prediction."

This work was carried out under the guidance of Mr. K. Suresh.

Slide 2: Outline

In this presentation I will discuss:

- Introduction
- Motivation
- Literature Survey
- Research Gaps
- Objectives
- Methodology
- Dataset
- Results
- Comparative Analysis
- Conclusion
- Future Scope

Slide 3-5: Introduction

Healthcare systems worldwide are facing increasing challenges due to chronic diseases such as:

- Diabetes
- Cardiovascular diseases
- Hypertension

- Thyroid disorders
- Kidney diseases

These diseases usually develop slowly and are often diagnosed at later stages.

Late diagnosis leads to:

- Increased treatment costs
- Higher risk of complications
- Reduced quality of life

Artificial Intelligence and Machine Learning provide an opportunity to analyze healthcare data efficiently and identify disease risks at an early stage.

Therefore, our research focuses on developing an AI-based framework for early disease prediction using Electronic Health Records.

Slide 6: AI in Healthcare

Artificial Intelligence can analyze large healthcare datasets and discover hidden disease patterns.

In our research:

- Random Forest performs disease prediction.
- Deep Learning techniques help understand complex relationships.
- SHAP provides explainability.

This makes the system both accurate and interpretable.

Slide 7: Motivation

Our motivation comes from four major challenges:

1. Increasing prevalence of chronic diseases worldwide.
2. Availability of huge healthcare datasets.
3. Existing systems mostly predict a single disease.
4. Lack of explainability in AI models.

Therefore, we aim to develop a multi-disease prediction system that is accurate, explainable, and clinically useful.

Literature Survey

Existing studies have explored:

Patel et al.

Used Explainable AI with EHR data.

Strength:
Interpretable predictions.

Limitation:
Performance depends heavily on data quality.

Denecke et al.

Used Transformer models.

Strength:
Captures complex relationships.

Limitation:
High computational cost.

Smith et al.

Used LSTM and RNN models.

Strength:
Good temporal analysis.

Limitation:
Requires large datasets.

Research Gaps

Despite existing advancements, several gaps remain:

1. Most systems predict only one disease.
2. Temporal health changes are often ignored.
3. Models act as black boxes.
4. Healthcare datasets contain missing and noisy data.
5. Existing systems provide predictions but not actionable recommendations.

Our work addresses these limitations.

Problem Statement

Accurate multi-disease prediction remains difficult because healthcare data contains complex nonlinear relationships.

Traditional approaches:

- Are disease specific
- Lack interpretability
- Cannot effectively model disease progression

Therefore, we propose an AI framework capable of:

- Predicting multiple diseases
- Providing explanations
- Supporting preventive healthcare

Objectives

The objectives of our research are:

1. Predict multiple chronic diseases simultaneously.
2. Use AI and Machine Learning techniques.
3. Improve prediction accuracy.
4. Provide explainable predictions using SHAP.
5. Support proactive healthcare decision making.

Dataset Description

We used a public healthcare dataset obtained from Kaggle.

Dataset details:

- Total Records: 15,029
- Training Samples: 11,460
- Testing Samples: 3,569
- Dataset Size: 1.2 GB
- Data Type: Electronic Health Records

The dataset contains:

- Demographic information
- Clinical measurements
- Lifestyle factors

Research Contribution

The major contribution of our work is:

Development of an AI-based multi-disease prediction system using:

- Random Forest
- Explainable AI (SHAP)

The system predicts disease risk and provides personalized preventive recommendations.

Proposed Model

Our methodology consists of three stages.

Stage 1: Data Preprocessing

- Data Collection
- Data Cleaning
- Missing Value Handling

- Normalization
- SMOTE for class balancing

Stage 2: Feature Selection & Modeling

Input Features:

- Age
- Gender
- Blood Pressure
- Glucose
- Cholesterol
- BMI
- Lifestyle Factors

Random Forest is used for classification.

Stage 3: Explainability

SHAP is used to explain:

- Why the model predicted risk
- Which features influenced the prediction most

Results & Discussion

The proposed model successfully predicts disease risks using patient health data.

The system:

- Analyzes user health information
- Generates risk assessments
- Produces preventive recommendations

The Random Forest model effectively captures nonlinear relationships among health indicators.

Comparative Study

We compared our model with traditional machine learning algorithms.

Model	Accuracy
Logistic Regression	85.2%
SVM	88.45%
KNN	86.7%
Decision Tree	84.3%
XGBoost	91.8%
Proposed Random Forest	92.5%

Our model achieved the highest accuracy.

Conclusion

To conclude:

- We developed an AI-based healthcare prediction system.
- The system predicts multiple chronic diseases.
- SHAP improves transparency.
- Random Forest provides high prediction accuracy.
- The framework supports proactive healthcare management.

Future Scope

Future improvements include:

- Wearable device integration
- Federated learning
- Multi-modal healthcare data
- Real-time monitoring
- Enhanced privacy and security

Thank you.

I am happy to answer any questions.